

Internship Task 1

Lab Setup & Foundations Report

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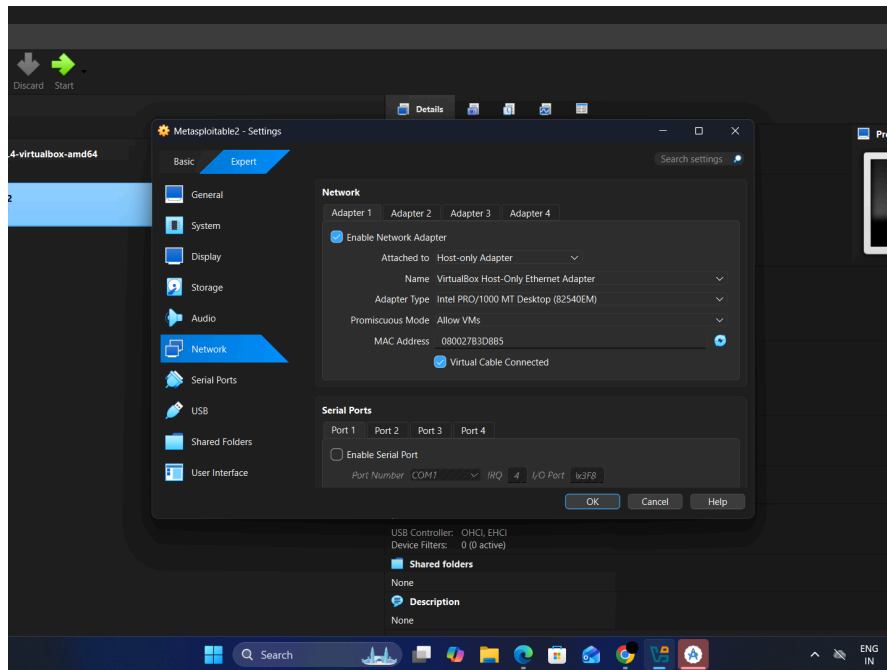
1. Objective

The primary objective of this task was to build a strong fundamental understanding of cybersecurity, networking, and cryptography while establishing a professional hacking lab environment.

2. Lab Environment Setup

To ensure a safe and isolated testing environment, I have configured the following virtualization setup:

Component	Description
Hypervisor	VirtualBox installed on the host machine.
Attacker Machine	Kali Linux (Latest Version).
Target Machine	Metasploitable2 (Intentionally vulnerable Linux VM).
Network Configuration	Configured a Host-Only Adapter to create a private network between the machines, preventing exposure to the external internet.

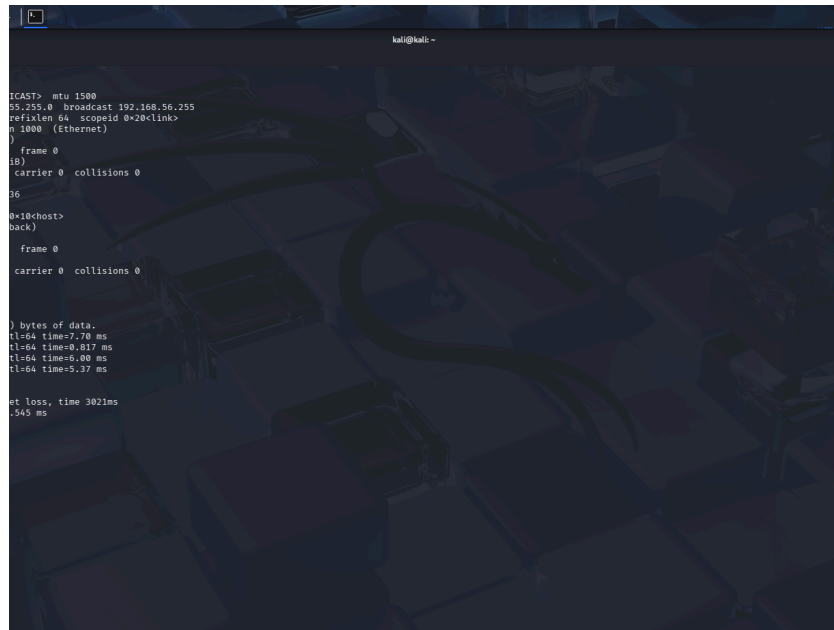


3. Technical Implementation & Tools

3.1 Networking Basics

I verified network connectivity using the following Linux commands:

- **ifconfig**: Used to identify the IP addresses of the Kali and Metasploitable machines.
- **ping**: Successfully verified a "heartbeat" connection between the attacker and target.
- **netstat**: Monitored active connections and listening ports.



3.2 Cryptography Hands-on

Using **OpenSSL**, I demonstrated the ability to secure data through symmetric encryption:

- **Encryption Command:** `openssl enc -aes-256-cbc -salt -in message.txt -out message.enc`.
- **Verification:** Confirmed that the original file was transformed into unreadable ciphertext.

4. Deliverables Summary

The following items have been prepared and are submitted as proof of successful task completion:

Deliverable	Status	Location/Description
Lab Setup Report	Complete	This document with verified screenshots.
GitHub Repository	Initialized	Initialized with a Linux cheat-sheet and project notes. 📄 File
Video Walkthrough	Complete	A 5-minute demonstration of the functional lab. 📄 File

